

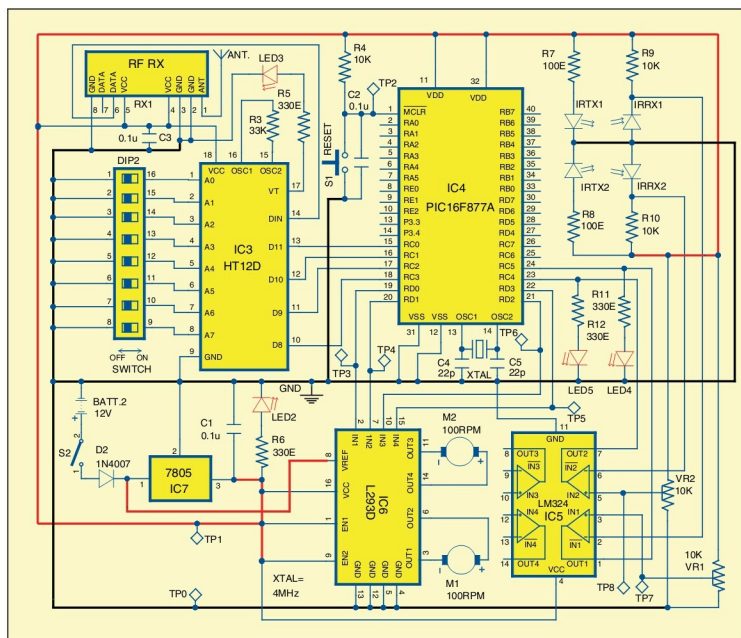


Fig. 4: Circuit of the robot

the encoder and the decoder through switches DIP1 and DIP2. The outputs from pins RD0 through RD3 of the microcontroller are fed to IN1 through IN4 of driver IC6 to drive motors M1 and M2, respectively. Enable pins EN1 (pin 1) and EN2 (pin 9) are kept high for always-enabled output.

Regulator IC7 is used for 5V regulated supply. 12V battery supply is directly connected to pin 8 of motor driver IC6. Two IR transmitter-receiver pairs are used for obstacle detection. Presets VR1 and VR2 are used to set the reference voltage for the two operational amplifiers in IC5, respectively.

When the robot reaches close to an obstacle, IR beam reflected off the

**TABLE III**  
**Test Points for Robot**

| Test point | Details                            |
|------------|------------------------------------|
| TP0        | Ground (GND)                       |
| TP1        | 5V                                 |
| TP2        | Reset                              |
| TP3        | Input for pin 2 of IC6             |
| TP4        | Input for pin 7 of IC6             |
| TP5        | Input for pin 10 of IC6            |
| TP6        | Input for pin 15 of IC6            |
| TP7        | Reference voltage for pin 3 of IC5 |
| TP8        | Reference voltage for pin 5 of IC5 |

obstacle is detected. Voltage at the inverting pins 2 and 6 of IC5 goes lower than the reference voltage and the output of both the operational amplifiers

becomes high. The same is indicated by LED4 and LED5.

Output from pins 1 and 7 of IC5 is fed to pins RC5 and RC4 of microcontroller IC4, respectively. Microcontroller controls the motor corresponding to the data received from the transmitter and obstacle-detection system. Any obstacle will block forward movement but motion in the other directions is still allowed.

Working of the circuit is straightforward. Hold the wireless steering in your hands as shown in Fig. 1 and adjust the tilt to obtain no-movement position. Now if you move the steering to the left and values from the accelerometer corresponding to that axis in-